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Mixtures of persistent organic pollutants are found in vital organs of late gestation human fetuses

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ABSTRACT

Persistent organic pollutants (POPs) are industrial chemicals with long half-lives. Early life exposure to POPs has been associated with adverse effects. Fetal exposure is typically estimated based on concentrations in maternal serum or placenta and little is known on the actual fetal exposure. We measured the concentrations of nine organochlorine pesticides (OCPs), ten polychlorinated biphenyl (PCB) congeners, and polybrominated diphenyl ether (PBDE) congeners by gas chromatography – tandem mass spectrometry in maternal serum, placenta, and fetal tissues (adipose tissue, liver, heart, lung and brain) in 20 pregnancies that ended in stillbirth (gestational weeks 36–41). The data were combined with our earlier data on perfluoroalkyl substances (PFASs) in the same cohort (Mamsen et al. 2019). HCB, p,p'-DDE, PCB 138 and PCB 153 were quantified in all samples of maternal serum, placenta and fetal tissues. All 22 POPs were detected in all fetal adipose tissue samples, even in cases where they could not be detected in maternal serum or placenta. Tissue:serum ratios were significantly higher in later gestations, male fetuses, and pregnancies with normal placental function. OCPs showed the highest tissue:serum ratios and PFAS the lowest. The highest chemical burden was found in adipose tissue and lowest in the brain. Overall, all studied human fetuses were intrinsically exposed to mixtures of POPs. Tissue:serum ratios were significantly modified by gestational age, fetal sex and placental function. Importantly, more chemicals were detected in fetal tissues compared to maternal serum and placenta, implying that these proxy samples may provide a misleading picture of actual fetal exposures.

Abbreviations: A, adipose tissue; B, brain; BMI, body mass index; GC-MS/MS, gas chromatography tandem mass spectrometry; GM, geometric mean; H, heart; HCB, hexachlorobenzene; HCH, hexachlorocyclohexane; IQR, interquartile range; IS, internal standard; Li, Liver; >LOQ, samples above the limit of quantification; <LOQ, samples below the limit of quantification; Lu, Lung; OCP, organochlorine pesticide; p,p'-DDE, dichlorodiphenyldichloroethylene; p,p'-DDT, dichlorodiphenyltrichloroethane; PCB, polychlorinated biphenyl; PBDE, polybrominated diphenyl ether; PeCB, pentachlorobenzene; PFAS, perfluorinated alkyl substance; PFDA, perfluorodecanoic acid; PFHxS, perfluorohexanesulfonic acid; PFNA, perfluorononanoic acid; PFOA, perfluorooctanoic acid; PFOS, perfluorooctanesulfonic acid; PFUnA, perfluoroundecanoic acid; POPs, persistent organic pollutants.

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1. Introduction

Living in a highly industrialized era, it is inevitable to be exposed to various chemicals in our everyday lives including persistent organic pollutants (POPs) such as organochlorine pesticides (OCPs), polychlorinated biphenyls (PCBs) and polybrominated diphenyl ethers (PBDEs). While these chemicals have been beneficial in pest and disease control and industrial processes, they have also been shown to have harmful effects on human health, wildlife and the environment (Bergman et al., 2013; Gore et al., 2015). These chemicals can be transported through air and water, thereby even affecting areas far from where they are used and released. Because this is a global concern, the United Nations Environment Program implemented the international treaty, Stockholm Convention on Persistent Organic Pollutants, to protect human health and environment from these chemicals by eliminating and restricting their production and use, and/or reducing unintentional release (Stockholm Convention, 2008). However, despite eliminating and restricting their production and use, these chemicals are still detected in biological samples because of their resistance to degradation, predisposition to bioaccumulate, and highly stable lipophilic properties (Bergman et al., 2013).

Fetal development is a critical window of vulnerability where the fetus is susceptible to environmental contaminants. Exposure to chemicals including POPs during this period has been of great concern due to possible disruption of development that may have significant short- or long-term health consequences for the unborn (Bergman et al., 2013; Gore et al., 2015). Typically, the connections between early life exposure and later adverse health outcomes in humans are studied in epidemiological pregnancy cohort studies. For example, early life exposure to POPs has been associated with low birth weight, epigenetic alterations, gestational diabetes, endocrine disruption, reduced sperm production, obesity and attention-deficit/hyperactivity disorder, among others (Callan et al., 2016; EFSA Panel on Contaminants in the Food Chain (CONTAM) et al., 2018; Jeong et al., 2018; Kim et al., 2018; Kobayashi et al., 2017; Lauritzen et al., 2018; Lenters et al., 2019; Lignell et al., 2013; Vafeiadi et al., 2014; Wikström et al., 2019). Because analysis of chemicals in the developing fetus is not possible in birth cohort studies, it is common to use different matrices such as maternal serum during pregnancy, placenta, and umbilical cord blood to estimate *in utero* exposure to chemicals (Frederiksen et al., 2009; Leonetti et al., 2016b; Müller et al., 2019; Nanes et al., 2014; Naqvi et al., 2018; Shu et al., 2018; Tanner et al., 2020). However, it is unknown whether these surrogate markers provide a good and reliable estimate of fetal exposure because data on actual concentrations of POPs as well as other environmental chemicals in human fetal tissues in relation to maternal serum and placenta samples is scarce. To the best of our knowledge, there are only two human studies available on fetal exposure to POPs and they focused on correlation between fetal liver and placenta (Doucet et al., 2009; Zota et al., 2018). In addition, it remains challenging to extrapolate toxicity from rodent studies to humans due to various reasons including toxicokinetics, where POP half-lives in rodents range from hours to days compared to years in humans (Huang et al., 2019).

Placenta was long considered as an organ that protects the fetus from harmful exposures. This notion was debunked by thalidomide, a pharmaceutical given to pregnant women in 1950s to counteract morning sickness and significantly affected fetal development causing malformations and even death (Vargesson, 2015). As environmental chemicals have been detected in cord blood and associated to altered fetal development, it is plausible to hypothesize that they reach the fetal tissues, potentially affecting their development and function although data demonstrating that is scant. Recently, it was shown that even ambient black carbon particles can be detected on the fetal side of the placenta, suggesting transfer of airborne particles over the placental barrier (Bové et al., 2019). We have also shown that perfluoroalkyl substances (PFAS) can be detected in human embryonic and fetal organs, indicating that the placenta indeed is penetrable for certain chemicals (Mamsen et al.,

2019).

Here, we wanted to study in detail the exposure of human fetuses to environmental pollutants in a wider context by measuring the concentrations of 22 persistent industrial chemicals in paired maternal serum, placenta, and fetal tissue samples of stillbirths (gestational weeks 36–41 weeks) of both sexes. The paired maternal and fetal tissue samples allowed an extensive evaluation of chemical exposure both in mother and the fetus. The well-characterized cohort gave us an opportunity to analyze factors that might affect chemical concentrations in the fetus. The results suggest extensive exposure of fetuses to industrial chemicals that is poorly predicted by maternal serum or placental samples. The data, which is fully disclosed, is of value to *in silico* modelling such as physiologically-based pharmacokinetics, and will aid in chemical health risk assessment during sensitive developmental windows.

2. Methods

2.1. Human fetal tissues, placenta and maternal blood samples

The samples used in this study were derived from different existing Swedish biobanks within the healthcare system. Therefore, we were not able to decide how, from what part of the organ, and how much sample was collected. Placenta (fetal side) and fetal tissues such as adipose, liver, heart, lungs, and brain had been deposited to the Perinatal Pathology Unit Biobank at the Karolinska University Hospital in Stockholm, Sweden, in the context of cause-of-death investigations of stillbirths. We obtained an ethical license to retrieve samples from this biobank for our studies covering cases investigated during years 2015–2016. We were able to connect most of the cases to maternal serum samples collected earlier during the same pregnancy in the Stockholm Medical Biobank. These serum samples were collected from routine first trimester prenatal visits to maternity care. Clinical data were gathered from autopsy reports and medical records, and the Stockholm classification of stillbirth (Varli et al., 2008) was used to identify the primary cause of death and certainty of diagnosis (Supp. Table 1). Subsequently, all data were completely anonymized.

2.2. Chemical analyses

Concentrations of nine organochlorine pesticides (OCP) [pentachlorobenzene (PeCB), hexachlorobenzene (HCB), isomers of hexachlorocyclohexane (α -HCH, β -HCH, γ -HCH), oxychlorodane, transnonachlor, dichlorodiphenyltrichloroethane (p,p'-DDT), dichlorodiphenyldichloroethylene (p,p'-DDE)], ten polychlorinated biphenyl (PCB) congeners (PCB 74, 99, 118, 138, 153, 156, 170, 180, 183 and 187) as well as three polybrominated diphenyl ether (PBDE 47, 99, and 153) were analyzed in the maternal serum, placenta, and fetal tissues, using accredited methods as described previously by Koponen et al. (2013) where all details including chemicals, consumables and instrumentation are given. Briefly, 200 μ L ethanol and 400 pg of 13 C-labelled internal standards of each compound in 100 μ L of toluene were added to serum (200 μ L) or fresh homogenized tissue (200 mg) samples in test tubes and mixed for 4 min at 1800 rpm to equilibrate internal standards to sample material. Dichloromethane-hexane (1:4) was added for extraction followed by activated silica to bind the sample water, ethanol and precipitate. Multilayer silica columns were used to clean the extracts and the eluate was concentrated for gas chromatography – tandem mass spectrometry (GC-MS/MS) analysis (Agilent 7010 GC-MS/MS system, Wilmington, DE, USA).

Included in each batch of samples ($n = 18$) were the control serum sample Standard Reference Material 1958 (SRM 1958) from National Institute of Standards and Technology (Gaithersburg, MD, USA), and an in-house low-concentration control sample (1–9 dilution of SRM 1958 with newborn calf serum). Average recovery from the SRM 1958 ranged from 88% to 111% of the certified concentrations and from 83% to 112% of the calculated concentrations from diluted SRM 1958. All of our

serum sample results from External Quality Assessment Scheme (EQAS) organized by Institut national de santé publique (Quebec City, Canada) for 6 OCPs, 10 PCBs, and 3 PBDEs have been acceptable ($|z| < 2$).

No certified material for POPs is available for human tissues. For this reason, we tested the feasibility of our original serum method for tissue samples with fish fillet sample from proficiency test arranged by European Union Reference Laboratory for Halogenated POPs in Feed and Food (<http://www.crl-dioxin-freiburg.eu/ptest.html>) that has assigned values for PCBs and PBDEs. Mean recovery of three parallel fish fillet samples ranged from 102 to 138% for PCBs (118, 138, 153, 156, 180) and 113–125% for PBDEs (47, 99, 153) of the assigned values depending on the congener. Example chromatograms of PCB 187 and PBDE 153 from placenta and liver tissues at around limits of quantification are found in [Supp Fig 1](#).

Concentrations in samples were reported as wet weight (pg/mL or pg/g). Limits of detection (LOD) ranged from 2 to 15 pg/mL or pg/g wet weight and limits of quantification (LOQ) from 5 to 40 pg/mL or pg/g wet weight. LOD and LOQ were calculated as the concentration corresponding to 3 and 10 times the standard deviation of the in-house low-concentration control sample, respectively, and rounded upwards to arrive at a conservative LOQ. For some analyses, we integrated the results with our earlier PFAS exposure assessment ([Mamsen et al., 2019](#)) for those fetuses that were included in both cohorts ($n = 13$).

2.3. Statistical methods

All statistical analyses were performed using the R programming language ([R Core Team, 2018](#)) through RStudio ([RStudio Team, 2016](#)) with the packages ggplot2 ([Wickham, 2009](#)), ggpvr ([Kassambara, 2017](#)), reshape2 ([Wickham, 2007](#)), dichromat ([Lumley, 2013](#)), RColorBrewer ([Neuwirth, 2014](#)), and ggpmisc ([Aphalo, 2019](#)). Analyses were performed only on samples above LOQ. All analyses were also carried out including samples above LOD. When all samples above LOD were included, the number of cases increased by 0–5 depending on the organ. This yielded similar results. Therefore, to have more reliable quantification, we decided to focus on samples > LOQ. Statistical differences between groups (i.e. male and females; normal placenta and with placental insufficiency) were tested with Wilcoxon rank-sum test. Correlation was investigated using Spearman's rank correlation coefficient. Tissue:serum ratios and tissue:placenta ratios were calculated only in cases where at least three paired samples were available using the formula:

$$\text{Tissue : serum ratio (\%)} = \frac{\text{Chemical concentration in tissue } \left(\frac{\text{pg}}{\text{g}}\right)}{\text{Chemical concentration in maternal serum } \left(\frac{\text{pg}}{\text{mL}}\right)} \times 100$$

$$\text{Tissue : placenta ratio (\%)} = \frac{\text{Chemical concentration in tissue } \left(\frac{\text{pg}}{\text{g}}\right)}{\text{Chemical concentration in placenta } \left(\frac{\text{pg}}{\text{g}}\right)} \times 100$$

Analyses of total chemical burden as well as fetal transfer among groups were focused on the 13 cases where data on OCP, PCB, PBDE and PFAS were available. The total chemical burden for different tissues were calculated by adding the medians of the tissue concentrations in the samples. To study relationships among chemical groups, the medians of the tissue:serum ratios of each chemical were used. Only the chemicals with at least three paired samples were included in the analyses. Kruskal-Wallis test was used to compare among chemical groups and Wilcoxon ranked test with Bonferroni correction for post hoc

comparisons. LogKow is the log of the octanol–water partition coefficient, which is used as an approximation of lipophilicity or hydrophobicity that could potentially predict how well the chemical is transferred from maternal serum to the fetal tissues when considering only passive diffusion processes. Medians of the log10 of the tissue:serum ratio were plotted as a function of logKow calculated from VEGA toolbox using standardized SMILES. The experimental value of logKow was primarily used, otherwise, the average of the predicted values from three models (Meyla/Kowwin, MLogKow, and ALogKow) were utilized. Spearman's rank correlation was done on OCPs/PCBs and PFASs. Significance was defined as $p < 0.05$.

3. Results

3.1. Sample characteristics

Twenty women aged 25–40 years (BMI 20.5–34.8 kg/m²) whose pregnancies resulted in stillbirth in late gestation (gestational weeks 36–41) were included in this study based on availability of samples. Maternal serum, fetal side of the placenta, fetal adipose tissue, heart, lung and brain samples were available for all 20 cases and liver was available for 17 cases. There were 10 male and 10 female fetuses included with an average fetal weight of 3128 ± 635 g. Causes of death were determined by a team of experts following the Stockholm classification of stillbirth ([Varli et al., 2008](#)). The most common causes of death were placental insufficiency/intrauterine growth restriction (55%) and infection (15%) with the diagnoses being certain in 45% of the cases ([Supp. Table 1](#)).

3.2. Mixtures of POPs are found in all maternal serum, placenta and fetal tissue samples

Twenty-two POPs were measured in all serum and tissue samples using liquid extraction - mass-spectrometric method ([Koponen et al., 2013](#)). Method is accredited for serum samples. An overview of the occurrence and concentrations of chemicals is shown as a heatmap in [Fig. 1](#) that is based on samples that were above the LOQ (i.e. concentration that can be determined with precision). We detected extensive mixture exposure in all sample types. All analyzed chemicals were detected in at least one sample type, and all sample types contained at least 15 chemicals. The most commonly detected chemicals were HCB, p,p'-DDE, PCB 138 and PCB 153, which were quantified in all serum,

placenta, and fetal tissue samples ([Fig. 1](#), [Suppl Tables 2 and 3](#)). The chemical present in the highest concentration per wet weight in all samples was p,p'-DDE, a metabolite of p,p'-DDT ([Fig. 1](#), [Suppl Tables 2 and 3](#)). The tissue with most chemicals quantified was fetal adipose tissue: every measured chemical was found in adipose tissue samples and the concentrations (pg/g wet weight) were clearly higher compared to other sample types ([Fig. 1](#), [Suppl Tables 2 and 3](#)). Importantly, the OCPs PeCB, α -HCH, γ -HCH, and oxychlordane were quantified in fetal tissues even when they were not detected in the maternal serum or placenta.

We next evaluated co-occurrence of the chemicals by calculating their correlations in maternal serum samples. Spearman's rank correlations showed that certain chemicals were significantly correlated. For example, most PCBs were significantly positively correlated with each other ([Supp Fig 2](#)). However, the OCPs detected in serum showed only

weak correlation to each other or to PCBs. These observations suggest that exposure to one PCB also predicts exposure to other PCBs while exposure to one OCP cannot be used to predict exposures to other OCPs.

We then wanted to know whether the concentrations of chemicals in maternal serum samples reflected the fetal tissue concentrations on a wet weight basis. We calculated Spearman correlations between paired serum and tissue samples. Some chemicals showed a significant positive correlation while others did not. For example, of the OCPs, p,p'-DDE showed significant positive correlation between maternal serum and all tissue samples, while HCB and transnonachlor did not (Fig. 2a). Significant positive correlations were found for most PCBs for all tissue types (Fig. 2a). When considering different tissue types, maternal serum concentrations correlated strongest with the concentrations in placenta as well as in fetal liver and brain (Fig. 2a).

In some studies, placental tissue is used as a surrogate when estimating fetal exposures (Frederiksen et al., 2009; Leonetti et al., 2016a; Nanes et al., 2014; Naqvi et al., 2018), and we therefore also considered correlations between paired placenta and fetal tissue samples. The placenta-fetus correlations were similar to maternal serum-fetus correlations. Significant positive correlations were found for most PCBs except PCBs 74, 156 and 183. β -HCH and p,p'-DDE were the OCPs with the strongest correlation (Fig. 2b). In general, concentrations in placenta correlated the best with concentrations in fetal brain (Fig. 2b).

In summary, these data show that all analyzed chemicals distribute from maternal circulation to fetal tissues, accumulating especially in the adipose tissue. Further, correlations between the proxy samples (i.e. maternal serum and placental tissue) with fetal organs were not found for all chemicals.

3.3. Chemicals accumulate in fetal tissues

Although correlation analyses inform about the relationships between chemicals in different samples, they do not reveal how the concentrations compare among sample types. To study the relationship in more detail on a wet weight basis, we calculated ratios between tissue concentrations (pg/g wet weight) and maternal serum concentrations (pg/mL) and expressed the results as %. These tissue:serum ratios were calculated only in those cases where the concentrations were above the LOQ and there were at least three paired samples available to allow for more accurate analysis. For all chemicals, the highest tissue:serum ratios were found in the adipose tissue, showing a wide range from 1,800% to 75,000% (Fig. 3a). This was followed by the liver (57%–22,400%), then heart (46%–3,200%) and lung (26%–1,430%) (Fig. 3a). The lowest tissue:serum ratios were found in placenta (46%–450%) and brain (26%–555%) (Fig. 3a). Because the maternal serum was taken during the first trimester while the placenta and fetal tissues were collected later at the time of fetal death, we also calculated the tissue:placenta ratios (%). A pattern similar to the tissue:serum ratios was found - highest in the adipose tissue, followed by liver, then heart and lungs, and lowest in the brain (Fig. 3b). In summary, most ratios were over 100% suggesting that on a wet weight basis, the chemicals accumulate to fetal tissues compared to maternal serum or placenta. We further note that there was a wide variation in the ratios, which could be due for example to inter-individual differences in chemical transport across placenta.

Altogether, our data show that the chemicals have a distinct distribution pattern to fetal organs. The wet weight concentrations in most fetal organs were clearly higher compared to placenta or maternal serum, extraordinarily high in fetal adipose tissue, and varied from individual to individual.

3.4. Adjustment of chemical concentrations by tissue lipid levels

The evaluated chemicals are notoriously persistent and lipophilic. Therefore, tissue lipid levels may explain the high wet weight concentrations of chemicals, for example, in adipose tissue. We wanted to investigate how the tissue levels might change if the chemical

concentrations were lipid-adjusted. Lipids are a broad group of biomolecules that are generally soluble in organic solvents but not in water, encompassing acylglycerols, waxes, phospholipids and sterols, among others. In the case of serum samples, cholesterol and triglycerides can be measured by commercial kits to calculate total lipids (Bernert et al., 2007). In tissues, the situation is more complex. Total tissue lipids can be roughly estimated by gravimetric methods based on extraction of lipids by solvents such as chloroform-methanol and weighing of the extracted lipid residue after solvent is evaporated (Lee et al., 1996). This approach requires more tissue than the Perinatal Pathology Biobank has available. The gravimetric method used to measure lipids in placenta ranges 1–5 g of tissue as starting material and estimated lipid content as 0.78–1.8% (Müller et al., 2019; Naqvi et al., 2018; Vizcaino et al., 2014). Similarly, Schecter et al. (2006) measured total lipids in human fetal livers. Amount of starting material was not reported. Average liver lipid content was estimated to be 2% in late gestation stillbirths. Zota et al. (2018) also employed the gravimetric methods to measure lipids in second trimester human fetal liver (2.6%) and placenta (1.1%). We did not find any other reports in the literature on lipid levels in human fetal organs. Moreover, it has been shown that human fetal and adult liver tissues differ in lipid composition (Kapitulnik et al., 1987), making extrapolation of lipids from adult tissues challenging. When chemical concentrations in liver and placenta were adjusted for reported lipid contents of 2% and 0.78–1.8%, respectively, the lipid-adjustment decreased the liver:placenta wet weight ratio by 10–60% (Supp Fig 3). For OCPs, the ratio decreased from 574% to 230–516% and for PCBs, from 293% to 117–260%. Similarly, the ratios would change for all tissue types if data on human fetal organ lipid levels were available. We provide the full chemical measurement data as a [supplement](#), enabling calculations based on lipid adjustment by any research group who may have human fetal organ lipid data at hand.

3.5. Gestational age, placental function and fetal sex affect chemical transfer to fetal organs

Our data suggest that the relationship between chemicals in maternal serum and fetal tissues may vary from case-to-case: for some organs, no clear correlations were found with maternal serum (Fig. 2) and there was variation in tissue:serum and tissue:placenta ratios (Fig. 3). To study what biological factors may affect the relationship, we focused on gestational age, placental function and fetal sex. The impact of these factors was assessed by studying differences in tissue:maternal serum ratios by gestational age, presence of placental insufficiency as primary cause of death, and fetal sex.

Although the time range in our study is narrow, from gestational weeks 36–41, we found a significant positive correlation of gestational age in days with several tissue:serum ratios (Fig. 4a). As POPs are persistent and bioaccumulative, their concentrations in fetal tissues can be expected to increase during gestation. There was a significant increase of transnonachlor in placenta, PCB 99 and 187 in adipose tissue and several PCBs and OCPs in brain. Hence, some of the variation in our data may be explained by differences in gestational age. Next, we compared the ratios between cases with normal placenta and cases with placental insufficiency. Cases diagnosed with placental insufficiency as primary cause of death had lower tissue:serum ratios compared to those without placental insufficiency (Fig. 4b). Significant differences were found for PCBs in adipose tissue, liver, heart and brain. Among OCPs, p,p'-DDE ratio in heart was higher in fetuses with normal placenta compared to those with placental insufficiency. Interestingly, the placenta:serum ratios were not modified by placental insufficiency. Finally, the effect of fetal sex was evaluated. We found that the tissue:serum ratios of several chemicals were significantly higher in male fetuses compared to females. Altogether, six of the studied 22 POPs (HCB, transnonachlor, p,p'-DDE, PCB 138, PCB 153, and PCB 180) showed sexually dimorphic exposure pattern (Fig. 4c). The highest differences were found in lung and heart followed by brain. The placenta:serum

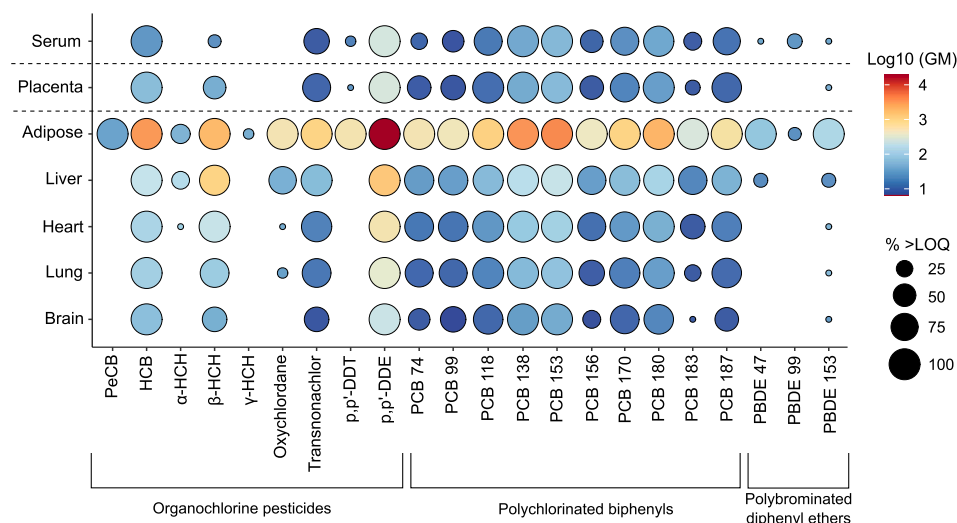


Fig. 1. Occurrence of persistent organic pollutants in pregnant mothers and their fetuses. An overview of the occurrence and wet weight concentrations of the 22 measured chemicals in maternal serum samples, placenta, and fetal tissues. The size of the dots indicates the quantification frequency *i.e.* the percent of samples that are above the limit of quantification (LOQ), and the color of the dots indicates log₁₀ of the geometric mean of the concentration (pg/mL serum or pg/g wet weight) measured in the samples. Absence of dots indicates that all samples were below LOQ. There were 20 samples for each maternal serum, placenta and fetal tissue and 17 liver samples.

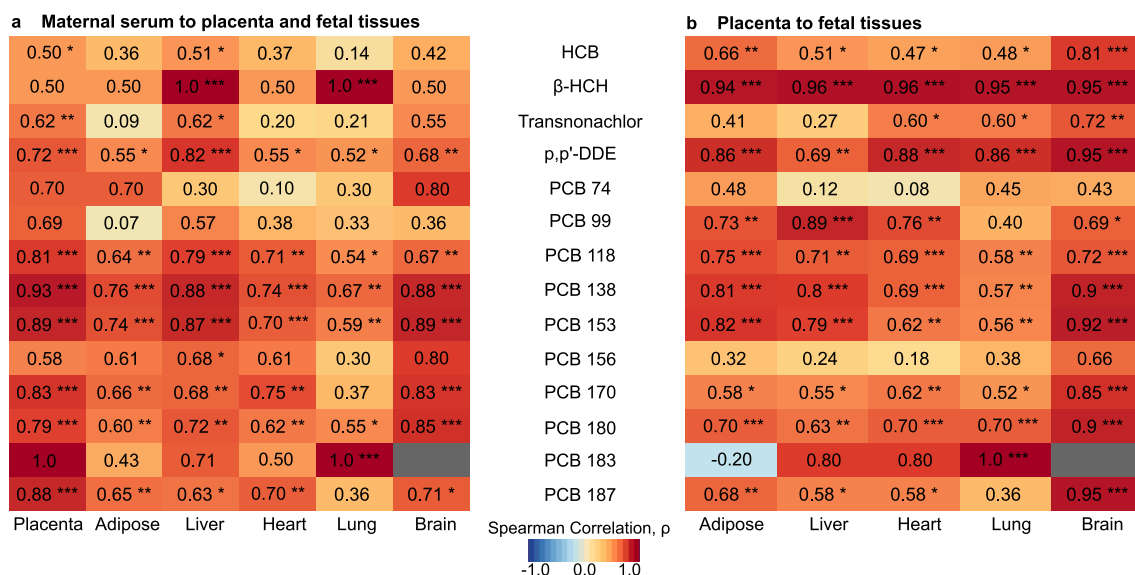


Fig. 2. Correlations between exposures in maternal serum, placenta and fetal tissues. Spearman's rank correlations were calculated to estimate relationships of chemical concentrations on a wet weight basis between maternal serum and fetal tissue samples (a) and placenta and fetal tissue samples (b). Data are presented as Spearman's rank correlation with ρ ranging from -1.0 (dark blue) to 1 (dark red) and significance marked as * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Correlations were calculated only in cases where paired samples were available in more than three cases. Otherwise, it was colored gray. Number of paired observations can be found in Fig. 3.

ratios were modified by fetal sex only in the case of p,p'-DDE. Because there were equal number of males and females for normal placenta and with placental insufficiency (and vice versa), and no significant differences in gestational age between males and females and between normal placental and placental insufficiency, these factors were affecting chemical transfer independently.

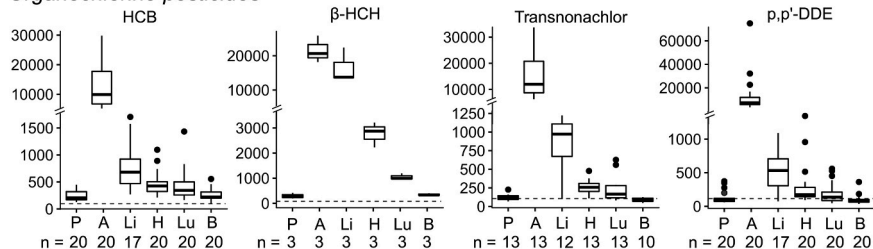
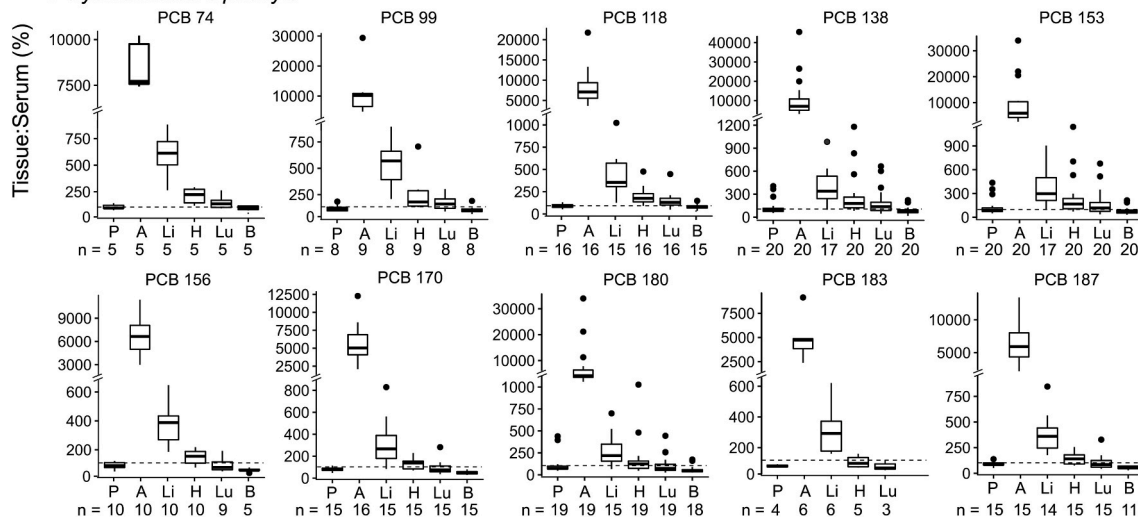
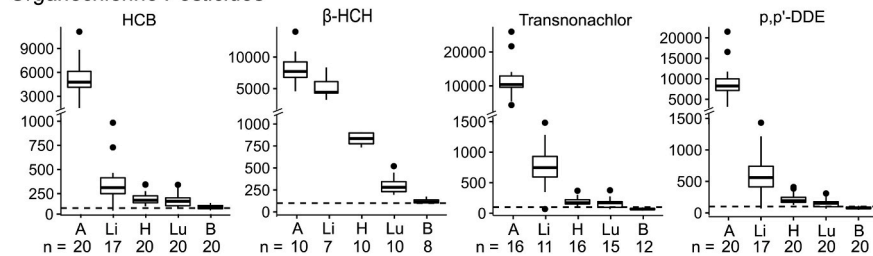
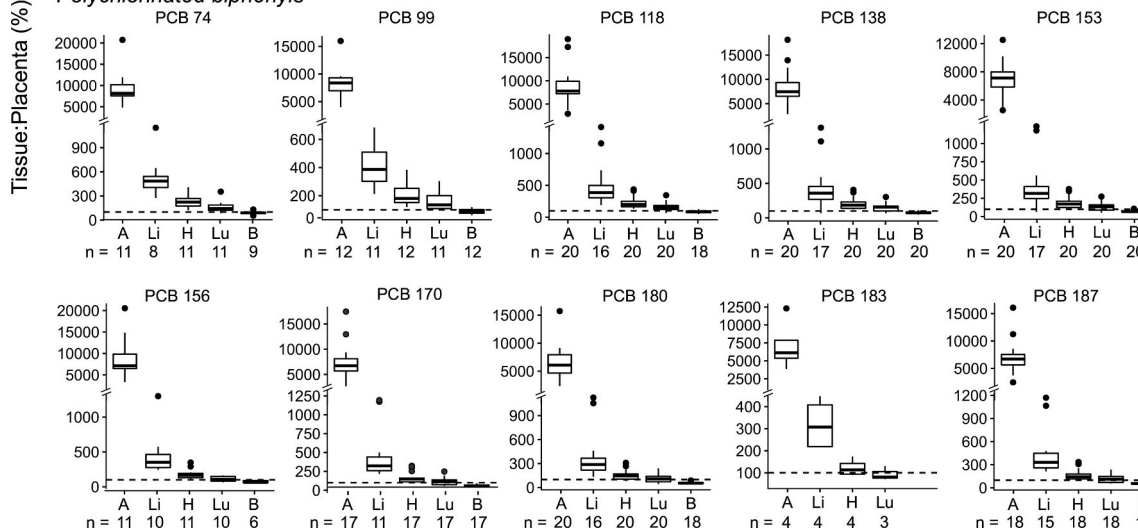
Altogether, these results demonstrate that several biological factors may affect the transfer of chemicals from mother to the fetus during *in utero* development causing inter-individual variation in fetal exposure concentrations. Longer gestation, male sex and normal placentation lead to intrinsically higher exposure concentrations than shorter gestation, female sex and placental insufficiency.

3.6. Total chemical burden is highest in the adipose tissue and lowest in the brain

OCPs, PCBs and PBDEs are likely not the only chemicals present in

the fetuses. While these types of POPs are lipophilic, there also exist other types of POPs such as perfluoroalkyl substances (PFAS) that are amphiphilic. Using our earlier data on fetal exposure to PFAS (Mamsen et al., 2019), we were able to calculate the total burden of exposure to 28 chemicals by tissue type in those 13 cases where data – including both previous and present data – for both chemical types were present. When all measured POPs and PFAS were considered, the highest chemical burden on a wet weight basis was found in the fetal adipose tissue where the sum of median concentrations reached 40,000 pg/g wet weight chemicals, whereas the brain had the lowest burden with the sum of median concentrations being 1,100 pg/g wet weight (Fig. 5a). OCPs and PCBs were predominantly found in the adipose tissue (46% p,p'-DDE, 9% PCB 153, 7% PCB 138 and 6% HCB) (Fig. 5b). On the other hand, in all other samples, more than half of the chemical burden was due to PFASs (Fig. 5b).

We evaluated if the tissue:serum ratios differ by chemical subgroup. PBDEs were not included because majority of the maternal serum

a Tissue:Serum*Organochlorine pesticides**Polychlorinated biphenyls***b Tissue:Placenta***Organochlorine Pesticides**Polychlorinated biphenyls*

(caption on next page)

Fig. 3. Chemicals have distinct accumulation patterns to fetal organs on wet weight basis. Rate of placental transfer to different fetal tissues was evaluated by calculating ratios between tissues and maternal serum samples. (A) Fetal tissue:maternal serum ratios were calculated by dividing the fetal tissue concentration by the maternal serum concentration and (B) fetal tissue:placenta ratios by dividing the fetal tissue concentration by the placenta concentration on a wet weight basis, and expressing the results as %. Dotted lines marks 100% at which the concentrations can be considered equivalent on wet weight basis. Majority of the ratios were clearly over 100% when using maternal serum or placenta as a reference, reflecting accumulation in fetal tissues on wet weight basis. The pattern was similar to all chemicals: adipose tissue showed the highest ratios followed by liver, heart, lungs and brain. Boxplots consist of the interquartile range (IQR) and the median, and whiskers are 1.5*IQR. Outliers are defined as values greater than 1.5*IQR and depicted with dots. Number of observations (n) is given in the graphs. There was less than three paired samples for brain:serum and brain:placenta ratios of PCB 183, hence, were not included in this graph. Abbreviations: P- placenta, A – adipose, Li – liver, H – heart, Lu – lung, B – brain.

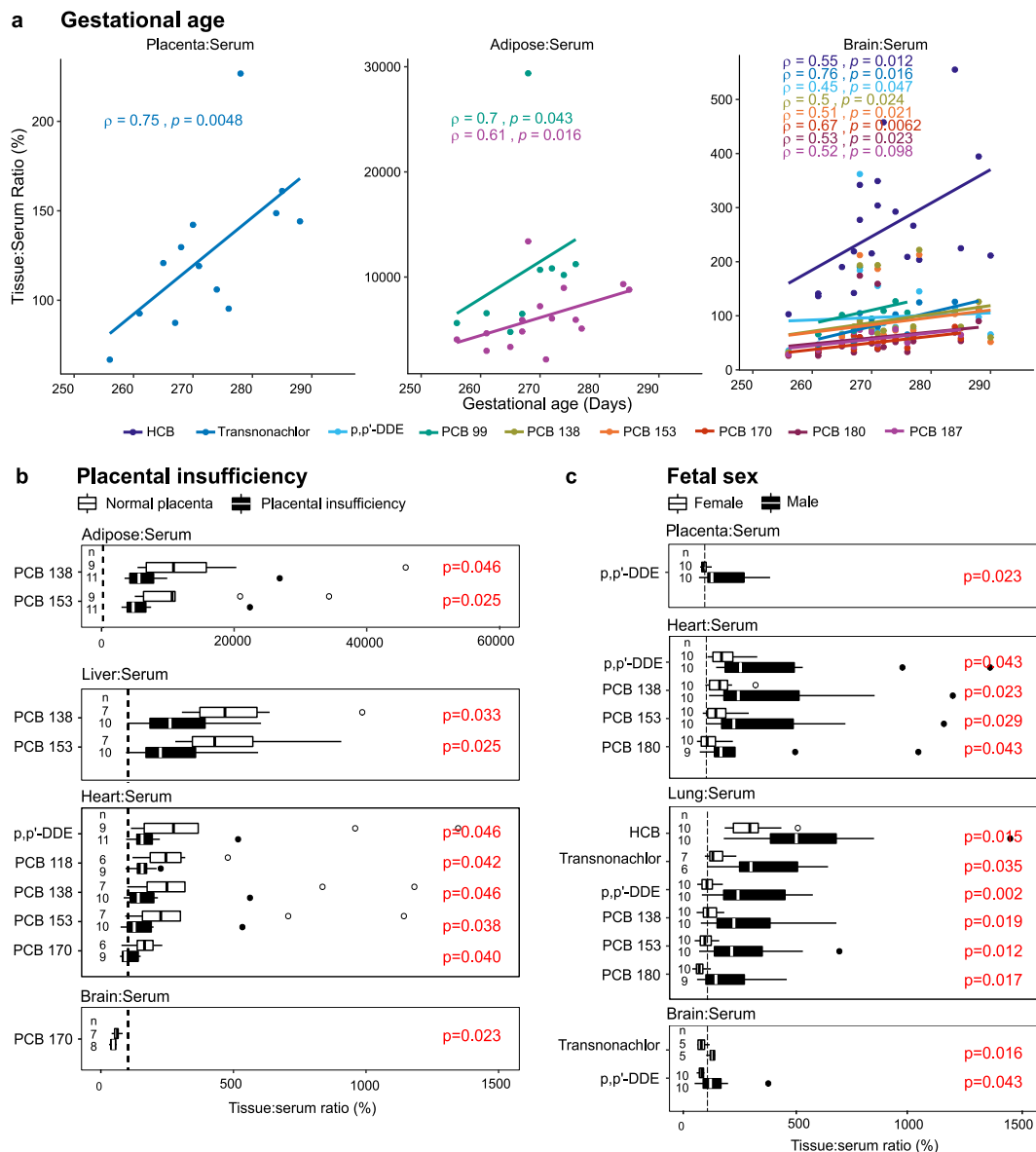


Fig. 4. Gestational age, placental function and fetal sex affect the relationship between maternal serum and fetal tissue chemical levels. To study biological factors affecting fetal exposure to chemicals, we tested the impact of gestational age, placental insufficiency and fetal sex on the tissue:serum ratios (%). The ratios were calculated by dividing the tissue wet weight concentrations by maternal serum concentrations and plotted as a function of gestational age (a), stratified by placenta insufficiency (b) or by fetal sex (c) for further analysis. a) There was accumulation of OCPs and PCBs in placenta, adipose tissue and brain as determined by Spearman correlation (ρ) between gestational age in days and tissue:serum ratios. b) Adipose, liver, heart and brain in pregnancies with normal placenta had higher ratios compared to those with placental insufficiency. c) Male fetuses had higher ratios of OCPs and PCBs in placenta, heart, lung and brain than female fetuses. For boxplots, dotted lines marks 100% at which the concentrations in placenta and fetal organs can be considered equivalent to maternal serum on a wet weight basis. Boxplots consist of the interquartile range (IQR) and the median, and whiskers are 1.5*IQR. Outliers are defined as values greater than 1.5*IQR and depicted with dots. Statistical differences between males and females and between with and without placental insufficiency were tested with Wilcoxon rank-sum test. Numbers of observations (n) are given in the figures.

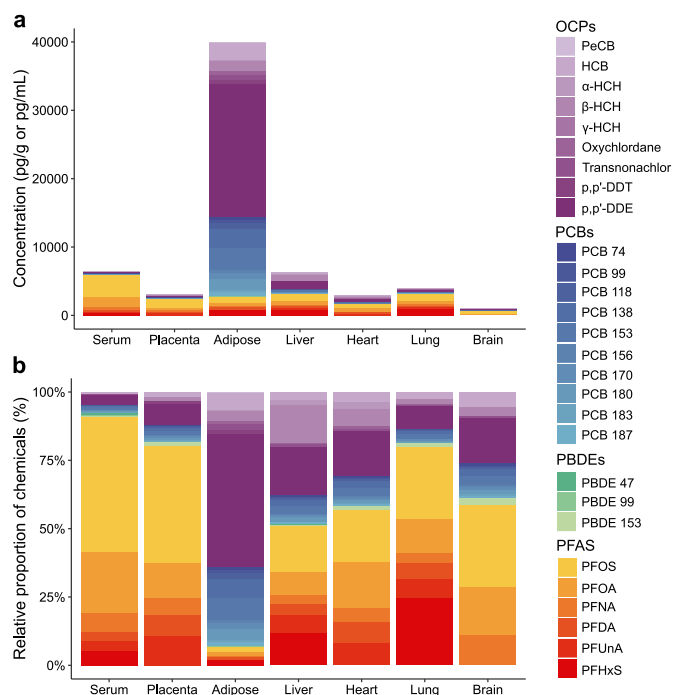


Fig. 5. Total chemical burden in fetal tissues. To evaluate the burden of chemicals in different tissues, we summed up the median exposures of 28 chemicals to arrive at cumulative POP burden (pg/mL or pg/g weight) in the different samples. a) The cumulative burden was highest in the fetal adipose tissue and lowest in the brain. b) The relative proportion of POPs in the samples was calculated by dividing the chemical medians by the cumulative POP burden in each organ. Majority of the chemicals in the adipose tissue were OCPs and PCBs. On the other hand, the chemicals in the other organs were mainly PFASs.

samples were below LOQ. A general trend was observed where OCPs had significantly higher tissue:serum ratios compared to PCBs and PFASs (Supp Fig. 4). To study whether the observed differences between chemical groups might be described by physicochemical properties, we considered LogKow values, which is a measure of lipophilicity or hydrophobicity and could potentially be a surrogate for how well the chemical is transferred from maternal serum to the fetal tissues through

passive diffusion (Peng et al., 2012; Zhao et al., 2016). For OCPs and PCBs, there was a significant negative correlation for log₁₀(tissue:maternal serum) of the placenta, adipose, liver, heart, lung and brain (Supp Fig. 5a). On the other hand, a significant positive correlation was seen for PFASs in the liver:serum ratio (Supp Fig. 5b).

In summary, POPs cross the placental barrier and distribute to various fetal organs at different efficiencies depending on the tissue and the type of chemical. All fetuses were intrinsically exposed to mixtures of toxic persistent industrial chemicals. Total chemical burden on a wet weight basis was lowest in the brain and highest in the adipose tissue. Overall, our study suggests that multiple factors affect chemical transfer from mother to fetus. In addition, our data show that chemicals in maternal serum and placenta, samples typically used in human birth cohort studies to estimate fetal exposure, may give a poor approximation of the diversity and levels of chemicals present in vital organs of human late gestation fetuses (Fig. 6).

4. Discussion

4.1. Mixtures of POPs are found in all maternal serum, placenta and fetal tissue samples

The present study evaluates for the first time a broad range of chemicals in multiple human fetal organs. In total, the concentrations of 22 POPs were measured in the adipose tissue, liver, heart, lung, and brain of 20 third trimester human fetuses and in associated placenta and maternal serum samples. This allowed an evaluation of actual fetal exposure and transfer of chemicals from mother to fetus as well as determine factors that may affect transfer efficiency. This information may be used for toxicokinetic modelling to predict the chemical burden of other POPs and their potential effect on the human fetus. The current study significantly expands on limited studies on the presence of POPs in human fetuses (Curley et al., 1969; Doucet et al., 2009; Nishimura et al., 1977; Pusiol et al., 2016; Schecter et al., 2006; Zota et al., 2018), especially in the context of studying the correlation between mother and fetus, which only two of the studies were able to explore with data from fetal liver and placenta (Doucet et al., 2009; Zota et al., 2018).

HCB, p,p'-DDE, PCB 138 and PCB 153 were the most common chemicals that were measured in all samples, which is consistent with previous studies on maternal serum and placenta concentrations of these chemicals (Fisher et al., 2016; Muller et al., 2019; Vafeiadi et al., 2014). However, PBDEs were mostly below the LOQ in our cohort, which is a clear difference to other studies where PBDEs were abundant (Müller

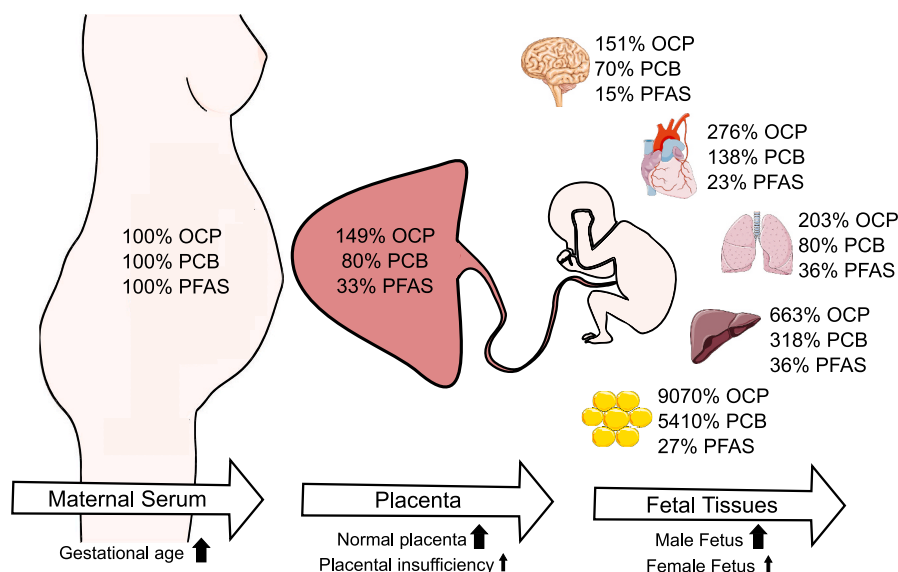


Fig. 6. Transfer of chemicals from maternal serum to placenta and fetal tissues. Summary of relationship between chemicals in maternal serum (pg/mL), placenta and fetal organs (pg/g) on wet weight basis. Biological factors that may affect the relationship (tissue:serum ratios) are marked with arrows (i.e. thicker arrow means higher concentration of chemicals). Highest transfer was seen for OCPs in the adipose tissue and lowest for PFAS in the brain. The ratios were positively affected by gestational age, normal placental function, and male fetus. The ratios are likely affected by tissue lipid levels, which we were not able to test systematically. Data shown are median serum:fetal tissue ratios of each chemical group. Polybrominated diphenyl ethers were not included because majority was below the limit of quantification.

et al., 2019; Oulhote et al., 2018). Doucet et al. (2009) found a general trend of PBDEs > OCPs > PCBs for both placenta and fetal liver in early pregnancy. Zota et al. (2018) also widely detected PBDEs in fetal liver in mid-pregnancy. Pusiol et al. (2016) detected OCPs in the fetal brain, mainly composed of chlordanes and p,p'-DDT. Moreover, the concentrations of OCPs, PCBs, and PBDEs found in the earlier cohorts (Curley et al., 1969; Nishimura et al., 1977; Schecter et al., 2006) of fetal organs were around 500-fold higher compared to our study. This may be due to the time period of sample collection of these cohorts (Curley et al., 1969; Nishimura et al., 1977), which were decades ago when OCPs and PCBs chemicals were not yet regulated. Moreover, the fetal tissues were obtained at different trimesters, suggesting that developmental stages may affect the concentrations. Another factor may be due to regional differences and more widespread use of PBDEs in North America compared to Sweden (Doucet et al., 2009; Schecter et al., 2006). Additionally, other factors also include history of local fire safety regulations, type of products in commerce, ethnicity and diet. Nonetheless, mitigation strategies through regulatory action may still work as seen by the decline of PBDE levels after being banned in California (Zota et al., 2013).

There was a positive correlation among the PCBs in maternal serum samples. However, this was not seen with the OCPs. PCBs are a family of 209 chemicals for which 'indicator PCBs' have been identified (i.e. PCBs 28, 52, 101, 118, 138, 153, and 180) to assess the exposure at large. PCBs typically occur as mixtures in food leading to co-exposure (Faroon et al., 2003). However, there is no indicator to estimate exposure to all OCPs due to lack of correlations between different types of OCPs, as we also detected in our cohort. Regardless of correlations between chemical groups, OCPs and PCBs were commonly detected in all samples, corroborating the "cocktail exposure" phenomenon which is well established by various biomonitoring programs in different countries (Porta et al., 2008). Our focus was on persistent contaminants because their concentrations will not be affected by sampling and storage conditions of biobank samples like the ones in our study. However, it is important to realize that the chemicals we focused on are far from the only ones in our environment. According to recent estimates, approximately 335,000 different chemicals are or have been in chemical commerce (Wang et al., 2020). Biomonitoring studies focus only on a small fraction of these chemicals, but regularly find several hundred man-made chemicals in human biological samples (Porta et al., 2008). This means that the old so-called 'legacy chemicals' like POPs persist in populations and are accompanied by an increasing number of newer chemicals. All populations are inherently exposed to multiple chemicals and our data shows that this applies even to fetuses. The fetuses had quantifiable concentrations of at least 15 of the 22 POPs analyzed and most likely these exposures are accompanied by multiple other chemicals that were not measured in this study. This calls for urgent development of better chemical health risk assessment, as the current approaches to evaluate chemical safety are based on assessing one chemical at the time (Demeneix and Slama, 2019; Drakvik et al., 2020).

4.2. Maternal serum and placenta may give misleading picture of actual fetal exposure

For the first time, we were able to correlate individual maternal serum concentrations of OCPs, PCBs, and PBDEs to corresponding placenta and fetal tissue concentrations. While some chemicals in serum (p,p'-DDE, PCB 118, PCB 138, PCB 153, and PCB 180) showed significant positive correlations with the placenta and all fetal organs on a wet weight basis, others lacked such correlation. In some cases, such as PCB 74, 99, 156, and 183, the poor correlations could depend on low numbers of observations because the correlations were calculated only when the concentrations were above LOQ in both serum and tissue. For some OCPs, PCBs and PBDEs, the correlations could not be calculated because they were mostly below LOQ in the maternal serum samples. Strikingly, these chemicals were still quantified in fetal tissues. This

means that the typical proxy samples of fetal exposure in birth cohort studies, maternal serum and placenta, would not give reasons to even consider PeCB, α -HCH, γ -HCH, and oxychlordane in fetal health although these chemicals might be present in fetal tissues. This suggests that maternal serum and placenta may give a misleading picture on the diversity and concentration of chemicals and are poor predictors of the actual fetal exposure. A similar result was reported by Doucet et al. (2009) based on 36 paired samples of placenta and fetal liver. They found that the OCPs, PCBs and PBDEs were several-fold higher in the fetal liver compared to the placenta, and the levels did not correlate. Zota et al. (2018) also demonstrated that fetal livers have higher PBDE concentrations compared to maternal serum. In summary, our results imply that actual fetal exposure concentrations may be hard to estimate based on maternal serum or placental concentrations because the diversity and concentrations of chemicals present in fetuses are not fully represented.

It is important to note that comparison of tissue concentration with the serum concentration is complex because of the different denominators (volume vs weight) and lack of data on lipid levels. Lipid standardization has been commonly discussed in the study of lipophilic compounds (Bernert et al., 2007). However, standardization for multiple matrices still warrants further examination. Zota et al. (2018) found no association between PBDE levels and lipids in maternal serum and placenta, suggesting that the assumption of equilibrium between total PBDE chemical exposure and lipid level may not be suitable. Moreover, lipid standardization has been shown to be highly prone to bias (Schisterman et al., 2005). This suggests further methodological studies are needed in exposure studies in multiple tissue types.

When we summed up the concentrations of 28 persistent chemicals in our cohort, the lowest chemical burden on a wet weight basis was found in the brain, which may be explained by the presence of the blood-brain barrier. During the third trimester, the blood-brain barrier has developed, thereby promoting a tighter control of chemical influx compared to other organs (Goasdou et al., 2017; Saunders et al., 2019). The highest chemical burden was found in the adipose tissue, which is in agreement with the lipophilic nature of most of these substances. The fetal adipose tissue develops towards the end of gestation (Symonds et al., 2017) and will provide the fetus with energy during the first days post-partum when lactation is not yet fully functional. Neonates mobilize their fat depot (15% of birth weight for full-term newborns) for energy metabolism before breastfeeding starts (Gustafsson, 2009). It is alarming that this depot of energy that sustains the baby during its first days of life contains such high concentrations of toxic chemicals. In addition, breast milk is an additional known source of POP exposure (Lignell et al., 2013; Muller et al., 2019). In summary, newborns face exposure to mixtures of persistent industrial chemicals derived from the mother.

4.3. Gestational age, fetal sex, and placental function affect transfer of chemicals from mother to fetus

We found several biological factors that significantly modified the transfer of chemicals from maternal circulation to fetal tissues. An association between fetal age and accumulation of OCPs and PCBs in the placenta, fetal adipose tissue and brain was observed. This is in line with the bioaccumulative and persistent nature of POPs, where absorption in organs exceeds the rate of metabolism and excretion (Bergman et al., 2013; Gore et al., 2015). However, we did not find accumulation in all fetal organs nor for all chemicals. This could be partly due to the short span of gestational age (i.e.: 36–41 weeks), covering only a part of the third trimester. It could also be partly explained by the lower statistical power due to smaller number of paired samples because we only used cases where data for both tissue and serum samples were above LOQ.

Another factor that significantly affected chemical transfer was placental insufficiency, which is a leading cause of fetal death (Varli et al., 2008) although also present in pregnancies that result in live birth (Spencer et al., 2019). In placental insufficiency, there is an increase in

vascular resistance, thereby limiting blood flow to the fetuses and decreasing transplacental transfer of oxygen and nutrients (Gagnon, 2003). In the same manner, transfer of chemicals to the fetuses may be then limited, which could explain the observed lower chemical transfer in fetuses with placental insufficiency. Placental insufficiency is often connected to fetal growth restriction and smaller birth weight, which is a predictor of later adverse health consequences. Chemical exposures are likewise often connected to lower birth weight and adverse developmental consequences. Our data suggest that there is an interaction between placental function and fetal chemical exposure, and encourage to take this into consideration in birth cohort studies.

Sexually dimorphic health effects of chemical exposures are sometimes reported, in particular in birth cohort studies. For example, associations between plasma OCP concentrations and reduced ponderal index (a measure of intrauterine growth) and birth weight were more pronounced in boys than girls (Callan et al., 2016). PBDEs were higher in placental tissues from pregnancies with male fetuses compared to female fetuses (Leonetti et al., 2016a). Moreover, negative associations between PBDEs in breastmilk and birth weight are seen in male infants, but not in females (Lignell et al., 2013). On the other hand, association of PFAS with lower birth weight (Wikström et al., 2019) and attention-deficit/hyperactivity disorder (Lenters et al., 2019) were found only in girls but not in boys. Interestingly, we discovered that chemical accumulation may be modified by fetal sex, being higher in male fetuses compared to female fetuses. Physiologically, these differences may be due to the differences in placental function between male and female fetuses, where there is higher placental vascular resistance in pregnancies with female fetuses compared to pregnancies with male fetuses (Rosenfeld, 2015; Widnes et al., 2017). This permits higher blood flow to male fetuses, exposing them to higher concentrations of chemicals. We hypothesize that this sex-specific distribution of chemicals may play a role in sexually dimorphic associations between maternal exposures and birth outcomes in previous studies (Callan et al., 2016; Lignell et al., 2013). Hence, fetal sex should be taken into account in birth cohort studies focusing on chemical exposure effects.

Exposure of fetuses to chemicals can be predicted using *in silico* tools. In fact, pharmaco- and toxicokinetic models of chemical distribution in humans are an essential part of *in silico* toxicity evaluation and drug development in general. These models rely on various physicochemical features of the compounds, and *in vitro* or *in vivo* data. Partitioning models which describe passive diffusion between systemic circulation and the various tissues could be approximated by the octanol-water partition coefficient (logKow) which is a commonly used surrogate to predict lipophilicity. In our data, there was a negative correlation between logKow and the tissue:serum ratios for the OCPs and PCBs. This indicates that for these chemicals, logKow alone cannot capture the expected disposition behavior. Hence, other physicochemical parameters or molecular descriptors would be required to establish and characterize their relationships. However, a positive correlation for liver:serum ratio was observed for PFASs, therefore, logKow seems to provide a reasonable estimate for the disposition behavior of these chemicals into this organ. Our data suggest that physicochemical properties are important since different chemical groups transfer at different efficiencies in different organs. Pesticides had higher concentrations in the fetal organs, followed by the PCBs, and PFASs. These differences may be attributed to molecular size, specific structural features, and chemical characteristics of each group. Further, this supports previous studies indicating that transport efficiency of the placenta can be related to the physicochemical properties of the substances (Chen et al., 2017; Yin et al., 2019; Zhang et al., 2021). These results encourage to study further the structure-activity relationship of these chemicals with a wide array of molecular descriptors. Furthermore, this emphasizes the importance of our data in understanding the intricacies of the complex kinetics of these chemicals in the fetus for an accurate physiologically-based pharmacokinetic model.

4.4. Limitations

Our study has several limitations. We focused only on a limited number of chemicals although it is well known that hundreds of thousands of chemicals are in commerce and likely end up in humans (Bergman et al., 2013; Gore et al., 2015; Wang et al., 2020). Our sample size is also modest. However, it is still the only study to date of POP exposure with an extensive set of human fetal organs paired with maternal serum and placenta. We were not able to adjust the chemical concentrations by tissue lipid levels, which could have provided more information to our analyses. Since our samples were retrieved from existing biobanks in the health care system, we were not able to choose how and when they were collected. The tissues were from fetuses that died *in utero*, which may not be representative of live born fetuses. However, this is the best approximation given that live birth samples are non-ethical. In addition, we had detailed autopsy reports that we could use to exclude cases with severe developmental anomalies. Another limitation is that the maternal serum was taken at an earlier time point (first trimester) compared to the tissues (third trimester). Even so, the tissue:serum ratios were comparable to the tissue:placenta ratios (where both tissue and placenta were taken at the same time point). Moreover, these chemicals are persistent and will not likely change drastically over a short period of time. Finally, the causes of stillbirth are sometimes hard to diagnose, and the most typical findings, such as placental insufficiency and growth restriction, affect live born infants as well. It is unclear why some pregnancies with these findings end in fetal death while others do not. Nonetheless, the causes of stillbirth were determined systematically by a group of experts following the Stockholm classification. We believe that the careful clinical characterization of our cohort adds value to the data and helps to generalize the findings to fetal exposure.

5. Conclusion

Overall, this study showed the quantification of OCPs, PCBs, and PBDEs in fetal tissues, implying that fetal exposure to POPs continues despite these chemicals being banned decades ago in Sweden. All studied chemicals were detected in the fetal adipose tissue even in cases when the chemical was not detected in the maternal serum and placenta. Serum:tissue levels were modified by gestational age, placental function, and fetal sex. Transplacental transfer of chemicals differed among groups of chemicals (OCP > PCB > PFAS). Overall, maternal serum and placenta may underestimate actual fetal exposure. These findings call for further evaluation of the current matrices used to estimate fetal exposure and establish a possible correction factor for a more accurate assessment of exposure *in utero*. We disclose the full data set on individual exposure concentrations to assist in building *in silico* models for prediction of human fetal exposure to chemicals.

Ethical approval

This project was approved by the Regional Ethical Review Board in Stockholm (EPN 2015/796–31/2).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2021.131125>.

Credit author statement

RDB: Investigation, Formal analysis, Writing – original draft, Visualization, Project administration, Data curation. M-TV: Investigation, Resources, Writing – review & editing; SG: Investigation, Resources, Writing – review & editing; LSM: Investigation, Resources, Writing – review & editing; CYA: Investigation, Resources, Writing – review & editing; NP: Investigation, Resources; DM: Resources. HK: Investigation; CHL: Investigation; PRA: Investigation, Writing – review & editing; PRU: Investigation, Writing – review & editing; PD: Conceptualization, Investigation, Writing – original draft, Supervision, Project administration, Funding acquisition. All authors approved the last version of the manuscript.

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Financial interest declaration

The financial supporters had no role in the study design, collection and analysis of data, data interpretation or in writing the report.

Data availability

Individual data supporting the findings of the study are available in the supplementary.

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